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Examples

Particle falling down vertically, bike moving in a straight road with acceleration.

2. The direction of motion alone can be changed without changing the magnitude (speed). If this happens continuously then we call it 'uniform circular motion'.
3. Both the direction and magnitude (speed) of velocity can be changed. If this happens non circular motion occurs. For example oscillation of a swing or simple pendulum, elliptical motion of planets around the Sun.

In this section we will deal with uniform circular motion and non-circular motion.

3.7.1 Centripetal force

If a particle is in uniform circular motion, there must be centripetal acceleration towards the center of the circle. If there is acceleration then there must be some force acting on it with respect to an inertial frame. This force is called centripetal force.

As we have seen in chapter 2, the centripetal acceleration of a particle in the circular motion is given by $a = \frac{v^2}{r}$ and it acts towards center of the circle. According to Newton's second law, the centripetal force is given by

$$F_{cp} = m a_{cp} = \frac{mv^2}{r}$$

The word Centripetal force means center seeking force.

In vector notation $\vec{F}_{cp} = -\frac{mv^2}{r}\hat{r}$

For uniform circular motion $\vec{F}_{cp} = -m\omega^2 r\,\hat{r}$

The direction $-\hat{r}$ points towards the center of the circle which is the direction of centripetal force as shown in Figure 3.38.

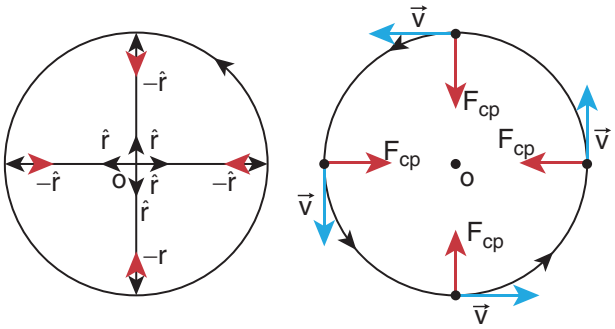


Figure 3.38 Centripetal force

It should be noted that 'centripetal force' is not other forces like gravitational force or spring force. It can be said as 'force towards center'. The origin of the centripetal force can be gravitational force, tension in the string, frictional force, Coulomb force etc. Any of these forces can act as a centripetal force.

1. In the case of whirling motion of a stone tied to a string, the centripetal force on the particle is provided by the tensional force on the string. In circular motion in an amusement park, the centripetal force is provided by the tension in the iron ropes.
2. In motion of satellites around the Earth, the centripetal force is given by Earth's gravitational force on the satellites. Newton's second law for satellite motion is

$$F = \text{earth's gravitational force} = \frac{mv^2}{r}$$

Where r- distance of the planet from the center of the Earth.